

Cloud Computing Assignment

Project: Architecting Scalable & Serverless Infrastructure

Objective

This assignment evaluates your ability to design and deploy cloud-native applications using modern architectural patterns like Microservices and Serverless.

1. Cloud Infrastructure & Networking

Design a Virtual Private Cloud (VPC) to host a multi-tier application.

- Configure **Public Subnets** for Load Balancers and **Private Subnets** for Application Servers.
- Set up a **NAT Gateway** to allow private instances to access the internet safely.
- Implement Security Groups (Firewalls) following the *Principle of Least Privilege*.

2. Compute & Auto-Scaling

Ensure the application can handle varying traffic loads.

- Deploy an **Auto Scaling Group (ASG)** that triggers based on CPU utilization or Request Count.
- Use an **Application Load Balancer (ALB)** to distribute traffic across multiple Availability Zones.
- Experiment with **Serverless Compute** (e.g., AWS Lambda or Google Cloud Functions) for event-driven tasks.

3. Storage & Database Management

Implement highly available storage solutions.

- Use **Object Storage** (S3/GCS) for static assets and set up a Content Delivery Network (CDN).
- Deploy a **Managed Relational Database** (RDS/Cloud SQL) with Multi-AZ deployment for failover.
- Utilize **NoSQL Databases** (DynamoDB/Firestore) for high-speed metadata storage.

4. Infrastructure as Code (IaC)

Automate the deployment process.

- Write a **Terraform** or **CloudFormation** script to provision the entire stack.
- Ensure all resources are tagged correctly for cost tracking and management.

5. Monitoring & Optimization

- Set up **CloudWatch** or **Stackdriver** dashboards to monitor latency and error rates.
- Define billing alerts to prevent budget overruns.